

FAST POWER SWITCHES FROM PICOSECOND TO NANOSECOND TIME SCALE AND THEIR APPLICATION TO PULSED POWER

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Abstract .

New methods of pulsed power are developed on the basis of new effects in high voltage semiconductor p-n junctions, discovered at Ioffe Physico-Technical Institute, Russia. These effects are:

- super fast voltage restoration [1];
- super fast reversible delayed breakdown [2].

This effects have been used to design two types of semiconductor devices:

1) Opening switches - Drift Step Recovery Devices (DSRD) with turn off times $0.5 \div 2.0$ ns at operating voltages 0.5 - 2.0 KV for one p-n junction.

2) Closing switches - Silicon Avalanche Shapers (SAS) with turn on times $50 \div 200$ ps at operating voltage 3-10 KV for one p-n junction.

DSRD may be considered as a solid state replacement for plasma opening switches and SAS may be considered as a replacement for overvoltaged spark gaps.

Both types of devices may be assembled in a stack (piled) with dozens of p-n junctions to increase operating voltages. Estimations show that operating voltage for the stack can reach 1 MV in the case of DSRD with turn off time ~ 1 ns and hundreds KV in the case of SAS with turn on times ~ 0.1 ns.

It is possible to switch all area of devices uniformly up to the limitations imposed by skin effect. Estimations show maximum switching current up to 10 KA for DSRD and > 2 KA for SAS.

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General physics of new switches operation.

2.1 Super fast recovery

When short forward current (pumping) pulse is applied to p-n junction (Fig.1) the plasma injection process starts. Highly enriched by carriers diffusion region appears near the p-n junction. The width of the region (W_d) is

$$W_d \approx \sqrt{D\tau_+}, \quad (1)$$

where D - diffusion coefficient, τ_+ - the pumping pulse length.

Part of the injected charge (equal to $\frac{\mu_p}{\mu_p + \mu_n}$,

where $\mu_{p,n}$ - carriers mobility) is accumulated in drift wave (Fig.2), which "head" propagates from diffusion region into the n-layer with the velocity (V_f)

$$V_f = j_+ / qN_d, \quad (2)$$

where j_+ - current density, N_d - n-layer doping.

When the current changes direction into the reverse, the drift wave moves back with the velocity, determined by (2) as well. The drift region is being compressed.

Near p-n junction in the diffusion region, the region of reverse concentration gradient appears. When concentra-

tion of electrons reaches zero level, neutrality condition can not be sustained longer and space charge region (SCR) arises. The rate of SCR voltage drop increase (dU/dt) is proportional to the velocity of the border of widening SCR (V_{SCR}).

$$dU/dt \approx \frac{qN_d l_{SCR} V_{SCR}}{2\epsilon} = E_m V_{SCR}, \quad (3)$$

where l_{SCR} - is the SCR width, ϵ - permittivity, E_m - maximum field at the p-n junction.

The velocity of SCR border moving into the highly reached diffusion region (DR)

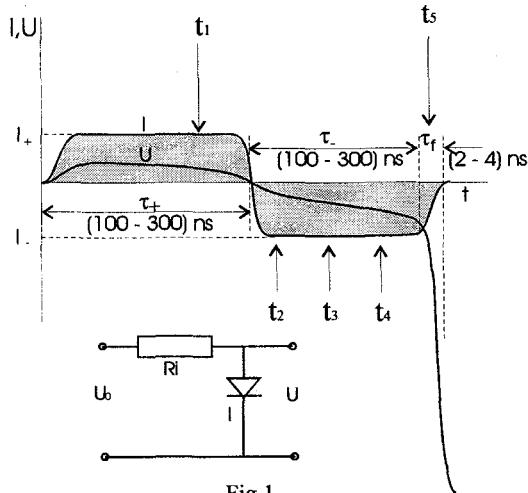


Fig.1
Fast recovery of high voltage p-n junctions

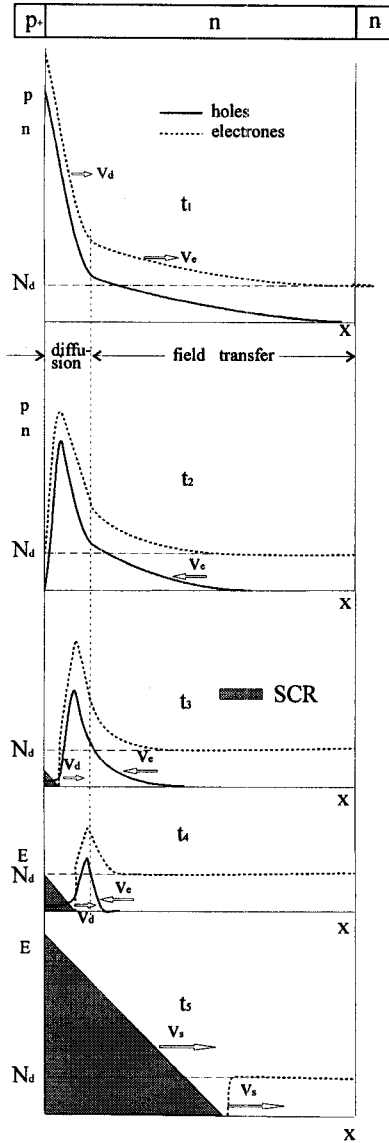


Fig.2
Carriers and field distribution during fast
recovery of high voltage p-n junctions

$$V_{SCR} \sim j_- / qn_m \quad (4)$$

is inversely proportional to carriers concentration in DR and is small due to high carriers concentration. Therefore the voltage rise rate is small as well (see moments $t_2 \div t_4$ on Fig.1 and 2). When the SCR border and drift wave head meet each other, diffusion region collapses. After the moment of collapse no nonequilibrium carriers exist in n -layer longer and conductivity current in SCR disappears. In accordance with (4) and (3), the SCR border velocity and dU/dt raises sharply. While diode voltage (U_d) is rising quickly, the diode current is going down quickly and reaches zero level, when U_d reaches the level of voltage source (Fig.1). At the current level

$j_- = j_s = qV_s N_d$, (where V_s is the saturated velocity of electrons) $V_{SCR} = V_s$. The maximum value of E_m is limited by breakdown value E_α . In accordance with (3) the maximum dU/dt for silicon diode is $U'_m = dU/dt \approx 2 \cdot 10^{12}$ v/s, when $V_s \approx 10^7$ cm/sec, $E_m \approx 2 \cdot 10^5$ v/cm.

It should be mentioned, that the evaluation, made above, may be applied to any diode, if condition $W_d \ll l_{SCR}$ is fulfilled. Turn off time (τ_{off}) of the diode with the maximum blocking voltage $U_m \approx \frac{E_\alpha l_m}{2}$ is

$$\tau_{off} \approx U'_m / 2U_m \approx \frac{l_m}{2V_s} \quad (5)$$

where l_m is maximum SCR width. Expression (5) yields nearly 1 ns for 1 KV diode, when $j = j_s$.

The diode recovery process, described above, involves no regenerative feedback and the current density distribution is stable and uniform across the diode area. So it is possible to increase breaked current I_m by the diode area (S) increase

$$I_m = j_s S. \quad (6)$$

It may be shown, that electrodynamics skinning limits for the turn off process is determined by the next relation

$$V_s / c < \frac{l_m}{R}, \quad (7)$$

where R is the radius of the diode, c - the electromagnetic wave velocity.

Therefore the maximum possible current I_m , breaked by diode, is determined by (5), (6), (7)

$$I_m \approx \epsilon E_m \tau_{off} C^2. \quad (8)$$

For silicon we have from (8) $I_m \approx 2 \cdot 10^4$ A, when $\tau_{off} \approx 1$ ns.

2.2

Fast reversible delayed breakdown.

When fast rising voltage ($U'_d = dU/dt > 10^{12}$ V/s) is applied to a diode biased in the blocking direction (Fig 3), the high density displacement current (j) appears in the space charge region. The field in the SCR increases and the border between SCR and neutral region (NR) moves with the velocity V_n , which is equal to the velocity of electrons in NR.

It may be shown that, when the condition

$$U'_d > U'_k = E_0 V_s, \quad (9)$$

(where $E_0 < E_b$ is the initial maximum field at p - n junction) is fulfilled, the field in NR exceeds the value E_s at which the velocity of electrons reaches the saturated value V_s .

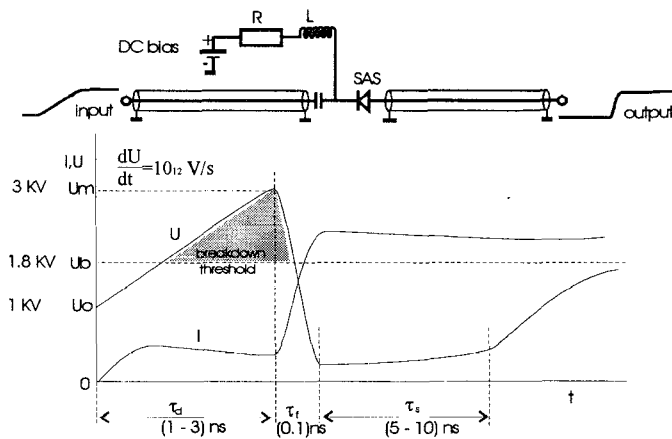


Fig.3

delayed breakdown in overvolted diodes

very high ionization rate due to very high field. Hole-electron plasma, born due to the ionization, sharply decreases

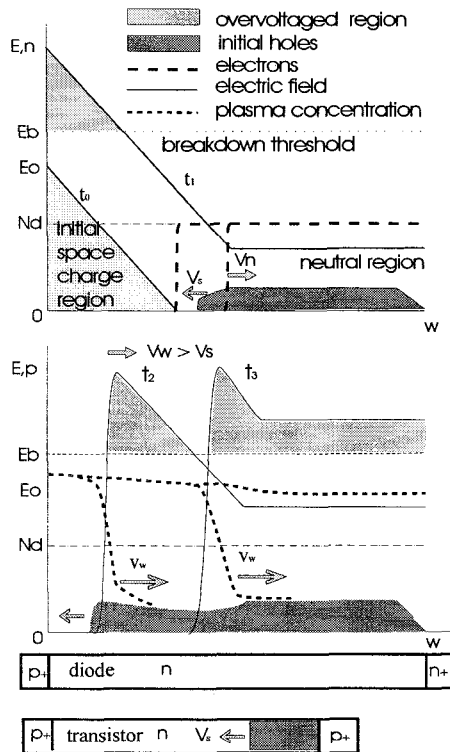


Fig.4
delayed ionization with fast
ionization wave excitation

the field in OR. The overvolted region is displaced toward the neutral region (Fig.4). Due to hole current from NR the extensive ionization starts again, field falls down, OR is displaced again and so on. The fast running wave appears. The wave velocity (V_w) may exceed the carriers saturated velocity V_s by orders of magnitude.

It may be shown that the plasma concentration (p_m) at the wake of the wave is

$$p_m \approx \frac{\epsilon \alpha(E_w) E_w}{q} \left(\frac{E_w}{b} \right), \quad (10)$$

where E_w is the maximum field at wave front, α - ionization coefficient, b - well known parameter in exponent for ionization coefficient (for silicon $b \approx 10^6$ v/cm).

So the ionization wave fills the volume of SCR by highly conducting plasma and turns on the diode. If SCR does not overlap entire n - layer, the overvolted region, when displaced into (NR), is uniform. Fast, due to high concentration of initial carriers ($n=N_d$), ionization starts in all NR simultaneously and entire NR is filled by plasma very quickly as SCR before has been.

The total measured turn on times for high voltage diodes are 50÷200 ps in 3-8 kV voltage range and strongly depends on dU/dt rate (the more the better). As follows from (9) the value of dU/dt needed to start super fast switching of diodes is close to the value of dU/dt for super fast voltage restoration, determined by (3). So these two effects ideally match each other. The fast voltage restoration at one diode may be used to switch on the other diode.

Generally speaking, the ionization wave flat front is unstable and current filamentation is possible. Neutral region allows to make front stable. In last case it is possible to increase diode current by increase of the diode area. Of course as in the voltage restoration case skin effect limits the size of the diode and the maximum current (I_m). It may be shown that

$$I_m < c^2 \epsilon E_{\alpha}^2 \tau_i / b, \quad (11)$$

When U_d is high enough, maximum field (E_m) at p - n junction can reach and exceed the static breakdown threshold (E_b). If diode leakage current is small, a number of initial carriers in SCR that could start ionization is small as well, and no ionization happens in SCR even when E is much higher than E_b (overvolted region (OR) appears).

In neutral region due to very high electron's concentration ($n = N_d$) the rate of impact ionization generation (G_{nr}) may be high in spite of relatively, to compare with SCR, low electric field

($G_{nr} \approx V_s \cdot \alpha_n \cdot N_d$, where α_n - ionization coefficient). Holes, born by ionization, drift into SCR (Fig.4), reach the overvolted region of SCR, where $E > E_b$ and give rise to

where E_α the field at the "head" of the wave, $\tau_i = \frac{1}{V_s \alpha(E_\alpha)}$ - ionization time, close to turn-on time.
For silicon we have from (11) $I_m \approx 2$ KA, for turn on time ≈ 100 ps.

3

New devices.

Both, fast restoration and delayed ionization effects are used in new semiconductor switches called drift step recovery devices (DSRD) and silicon avalanche shapers (SAS).

Each type of the devices may have one, two three $p-n$ junctions (diodes, transistors, thyristors). Additional $p-n$ junction adds to the devices new features, for example, better efficiency for long pulses, less voltage drop in turn on state, simplified "pumping" circuit and high (megahertz) pulse repetition rate for DSRD and so on.

In the case of SAS (Fig.5), for example, in transistor like structure (p^+np^+) (Fig.4), the initial holes may be injected from $p-n$ junction. Double injection in SAS-dynistor ($n^+p^+np^+n^+$) after its turn on could keep it in turn on state infinitely until the current is broken by external source.

Both types of switches mostly used as two terminal devices (diode like) and they can be assembled in a stack (piled) with many (dozens) $p-n$ junctions to increase operating voltage of such combined device. The end user sees the stack as single high voltage device (Fig.6).

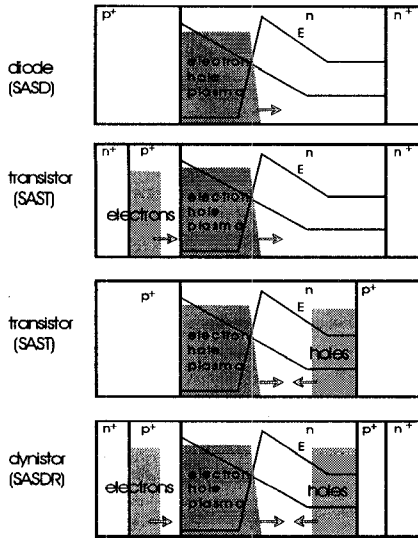


Fig.5
devices on delayed ionization

The global limitation for a number of devices in a stack is determined by the condition: the time of electromagnetic wave propagation along the stack (τ_w) should be shorter than switching time (τ_i). This condition leads to

$$U_m < c E_\alpha \tau_i / 2 \quad (12)$$

where U_m - maximum stack voltage, E_α - maximum field.

For silicon DSRD with 1 ns turn off time we have $U_m \leq 1$ MV, and for SAS with 0,1 ns $U_m \approx 200$ KV (the higher E_α for SAS has been taken into an account).

It should be noted, that in the case of two terminal DSRD stack, the pumping current and the reverse current are the same for each diode. So, if the losses of injected carriers are small (lifetime of carriers may be

as high as 100 μ s many orders of magnitude more than pumping time) the moments when current is broken are the same for each diode, i.e. all diodes in the stack are synchronized. In experiments we broke current for < 1 ns by the stack blocking 60 KV and consisting of nearly 100 diodes.

If one of the SAS in a stack turns on earlier than others, the SAS displaces voltage from itself to others. Due to additional voltage increase others SAS turn on earlier and they catch up one switched the first (self synchronization effect). In experiments we have switched on SAS's stack consisting of 6 SAS with turn on voltage > 25 KV for less than 100 ps.

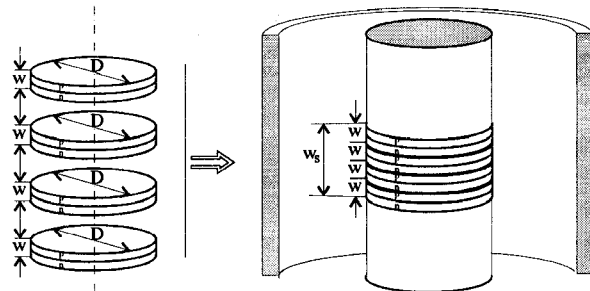


Fig.6
High voltage stack consisting of many p-n junctions

New devices may be used for pulse generation. SAS may be used in slightly modified well known gas discharge peaking circuits. The use of DSRD is different from the use of plasma opening switches. One of the most efficient and simple circuits is symmetrical double LC circuit (Fig.7)

Let us consider two similar LC- circuits with DSRD inserted in common branch of the circuits (Fig.7). Initially energy storage capacitors C1, C2 are charged, (the polarity is shown on Fig. 7) and switches S1, S2 are opened. When the switch S1 is closed, capacitor C1 discharges via inductor L1 and the diode (DSRD). The discharge current is forward current for DSRD, the resistance of DSRD is low and the current (I1) in C1, S1, L1,

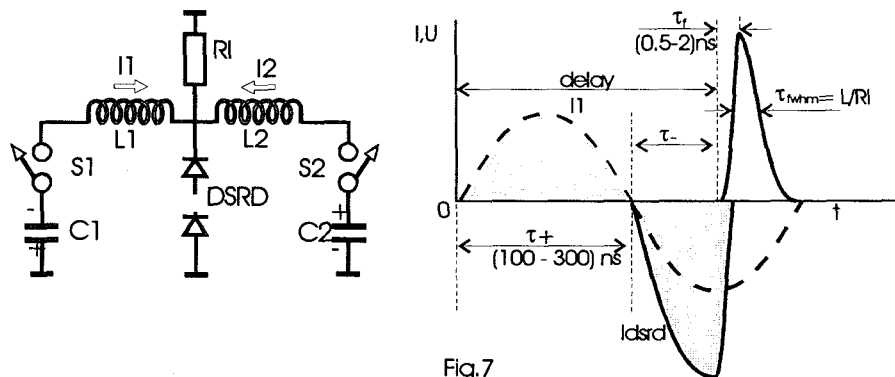


Fig.7

Power compressing cell on DSRD

DSRD circuit oscillates.

Half-cycle period of LC circuit must not be more than several hundreds nanoseconds. Minority carriers lifetime in DSRD is large - ten's and more microseconds. The total amount of electron-hole pairs, injected ("pumped") in DSRD during the first forward halfperiod of current oscillation is equal to the charge passed through the diode.

When the current changes its direction (the second halfperiod) the diode remains in high conducting

state due to stored electron-hole pairs. If at the moment, when current I1 crosses the zero -level ($t = \tau_+$), the second switch S2 closes, the C2 discharge current (I2) is added to the current of L1, C1 circuit. The total DSRD's current is doubled. At the moment of the current maximum the charge extracted for τ_- period of time from DSRD is equal to one injected for τ_- period and DSRD total current sharply (for 1-2 nanoseconds) is broken. At the moment all energy, initially stored in C1, C2 capacitors, are accumulated in L1, L2 inductors, and their currents reach maximum values. During the time of DSRD switch off processes its current is being switched into the load resistor RL. The front of the load current pulse is determined by turn off time of the DSRD, and the decay is $\sim L/RL$ (where L is the total inductance of L1 and L2 connected in parallel). The peak load voltage $U_{lm} \sim RL \cdot (I1 + I2)$ may be many times higher (> 10 times) than the initial capacitor voltage U_0 .

Such voltage multiplication is one of the important advantages of circuits based on DSRD. High voltage exists in circuit only very short time - nanoseconds, therefore high voltage corona and arc discharges troubles are not so severe as in the case of DC.

It should be mentioned, that total delay of shaped pulse is equal to 3/4 of LC circuit period and may reach hundreds nanosecond. The stability of the delay is determined by stability of LC circuit, which, as well known, may be better than 10^{-4} , so jitter of the delay may be less than 100 ps. When C1 and C2 are charged from one power supply, unstability of the supply does not influence on the moment of DSRD current break (the ratio of injecting current to extracting current does not depend on power supply voltage).

The circuit in the case of the ideal primary switches S1, S2 shown above is very simple and very efficient. Unideality of parts determined by loss of energy in L, C, the primary switches S1 and S2 may strongly influence on the efficiency of pulse generation.

Influence of losses may be defined by Q-factor of C,S,L, DSRD circuit or by the ratio of first current maximum at the first half-cycle to the second, when only one switch S1 is turned on. It turned out that the task of getting good total Q-factor (> 3) for high currents > 100 A is very hard.

It is very easy to get large Q-factor (> 100) for small (< 1 microhenry) iron less linear inductors even at high (> 1000 A) currents. Q-factor of low inductive capacitors with stored energy $D > 10^{-3}$ J is not so large as for inductors and for ceramic best capacitors is $\sim 10-30$.

The losses in switches play major role on Q-factor. There are two types of losses: transient losses - during turn-on processes and losses due to resistance in turn-on state. Actually they decreases Q factor of LC circuits down to the factor of two-five.

Due to low Q-factor symmetry of the circuit (Fig.7) is broken. With equal L and C in both circuits, the second halfperiod L1, C1 current I1 is not equal to the current I2 during the first halfperiod L2, C2 . When the DSRD breaks current, part of L2 current (I2) goes to the first inductor L1 additionally increasing total losses. Adjustment of parameters of the first LC circuit (increase of L1 and decrease of C1) can decrease the current interchange between L1 and L2, but, for example, if Q-factor is less than 3, overall efficiency of pulse forming circuit will still will be worse than 70-80%.

Every kind of known closing switch such as transistors (fields effect and bipolar), thyristors and dynistors may be used as a primary switch S1 and S2.

References

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